Neuroformer: Multimodal, Multitask Generative Pretraining for Brain Data

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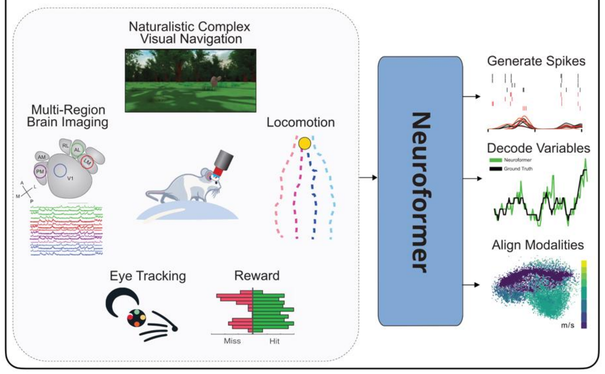
Introduction

Systems neuroscience experiments collect a vast amount of valuable data.

Currently there is no way to leverage this data.

We develop a framework to enable generative pretraining on diverse neural datasets.



Model

Framework:

- Reframe neuron IDs as vector embeddings ("tokens").
- Align multiple modalities using contrastive learning.
- Model neural decoding as a sequential autoregressive process.
- Model neural decoding as a dynamical system [2].

Workflow:

- Iteratively fuse the neural state with other modalities using a transformer that unrolls recurrently in space [3].
- Decode using a causal transformer decoder with two projection outputs: temporal prediction and neuron ID.

Optimization:

- Alignment (contrastive objective) [2]:

$$s(F,I) = g_f(F_{p,c})^T g_i(I_c) \qquad s(I,F) = g_i(I_c)^T g_f(F_{p,c})$$

$$p_m^{fi} = \frac{\exp(s(f,i_m)/\tau)}{\sum_{m=1}^{M}\exp(s(f,i_m)/\tau)} \qquad p_m^{if} = \frac{\exp(s(i_m,f)/\tau)}{\sum_{m=1}^{M}\exp(s(i_m,f)/\tau)}$$

$$L_{vnc} = \frac{1}{2}\mathbb{E}_{(F,I)\in d}[H(\boldsymbol{y}^{fi}(F),\boldsymbol{p}^{fi}(F)) + \boldsymbol{y}^{if}(I),\boldsymbol{p}^{if}(I))]$$

- Spatiotemporal decoding (MLE):

$$L = (\gamma)L_{vnc} + (\mu)L_{ce(I)} + (1-\gamma-\mu)L_{ce(dt)}$$

[1] Learnable latent embeddings for joint behavioral and neural analysis
Steffen Schneider, Jin Hwa Lee, Mackenzie Weygandt Mathis

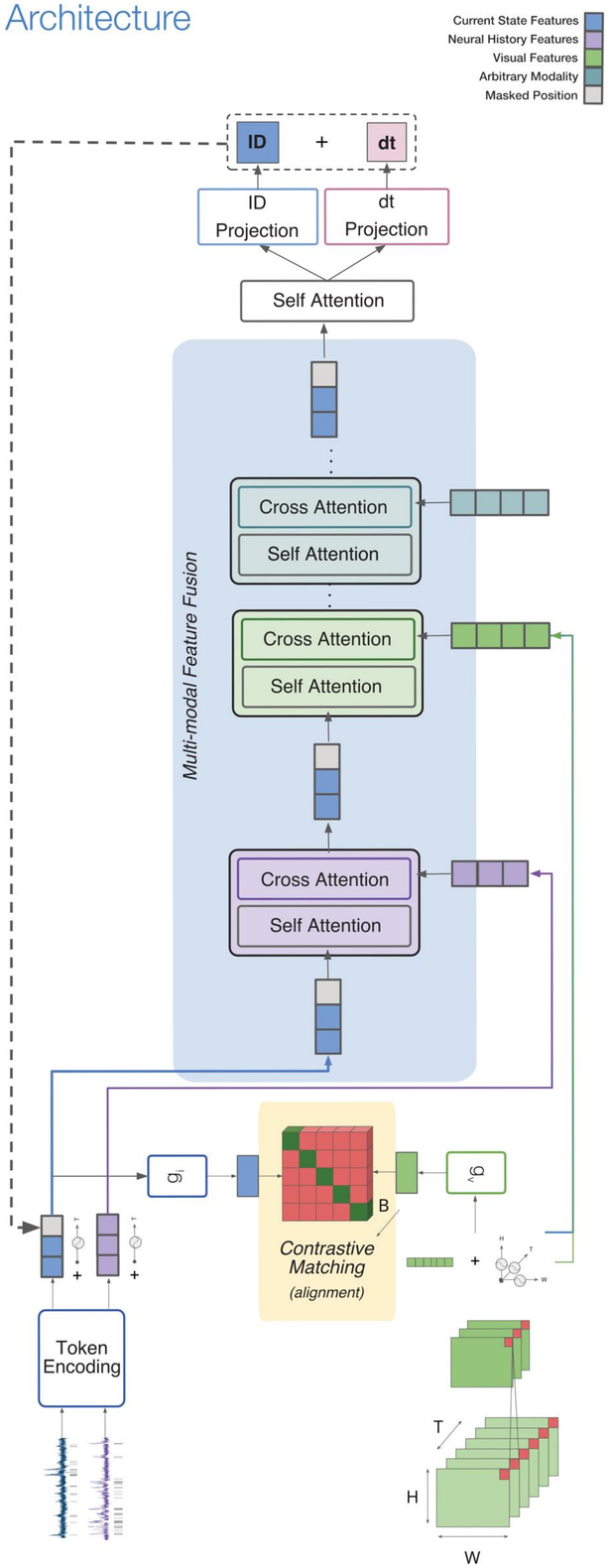
[2] LFADS - Latent Factor Analysis via Dynamical Systems
David Sussillo, Rafal Jozefowicz, L. F. Abbott, Chethan Pandarinath

[3] Perceiver: General Perception with Iterative Attention
Andrew Jaegle, Felix Gimeno, Andrew Brock, Andrew Zisserman, Oriol Vinyals, Joao Carreira

[4] The functional specialization of visual cortex emerges from training parallel pathways with self-supervised predictive learning
Shahab Bakhtiari, Patrick Mineault, Tim Lillicrap, Christopher C. Pack, Blake A. Richards

[5] Attention Approximates Sparse Distributed Memory
Trenton Bricken, Cengiz Pehlevanv

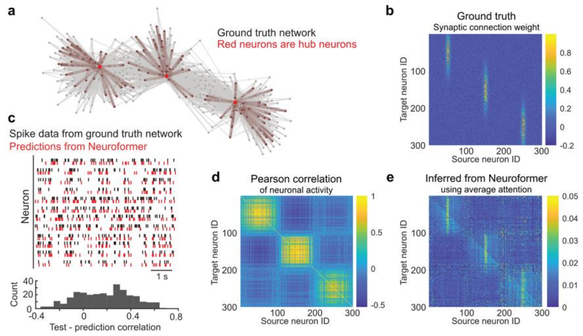
Architecture



Experiments

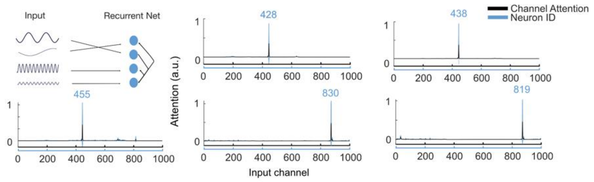
Self-supervised Learning of Neuronal Connectivity

- Simulated dataset of hub neurons.
- Self-attention uncovers ground-truth connectivity.
- Explains 20% variability (compared to 13% for Pearson correlation.)



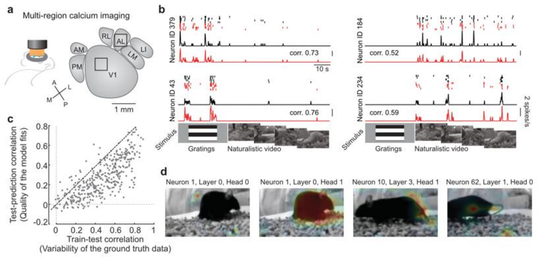
Recovering Cross-modal Relationships

- Simulated dataset of 1000 neurons, each receiving one of 1000 different periodic inputs.
- Cross-attention reveals the 1-to-1 input mapping.



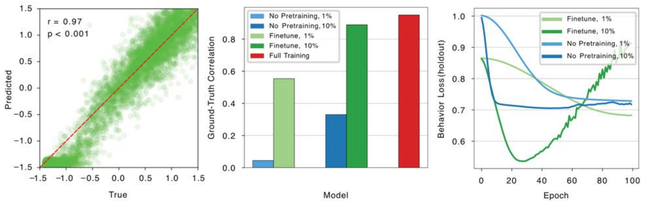
Long-horizon Simulations of Brain Activity

- Mouse watching a naturalistic video.
- Neuroformer can generate high-precision simulations of ground-truth trials of 32 seconds.
- Cross-attention between neurons and video stimulus reveal salient features.



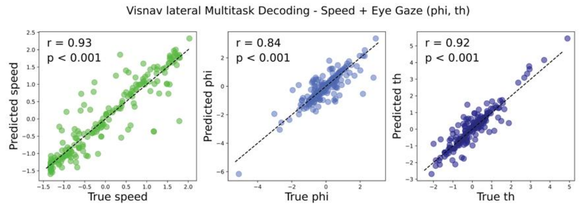
Fewshot Prediction of Mouse Behavior

- Pretrained models utilizing fewshot finetuning could accurately decode mouse speed.

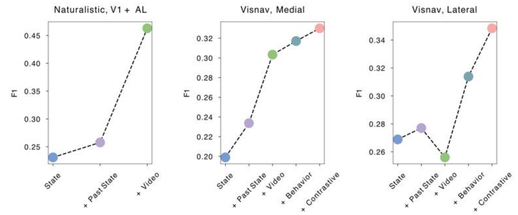


Joint Multitask Prediction

- Jointly predict eye position and speed.

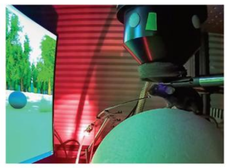


Alignment and Multimodality Significantly Improve Performance



Future

- Unify Diverse neural datasets to massively scale training.
- Joint neural training to improve other models.
- Incorporate within BCI applications.



Summary

- Neuroformer is a **Multimodal LLM** that leverages **brain data**.
- Neuroformer can generate high-precision simulations of neural data in a similar style to generative language models (e.g., GPT).
- When trained using a self-supervised objective, the model can intrinsically learn causal features such as connectivity and multimodal relations [5].
- We are now scaling training to explore the potential of brain data to improve models (*vision, language and robotics)*

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